

CENTRIFUGAL PUMP DESIGN CONSTRAINTS

SOURCE API Standard 610	EDITION 13th Edition	DATE June 2026
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Purpose. Provide an engineer-friendly design review guide that links each principal API 610 constraint to an application scenario, engineering consideration, and verification evidence.

Use limitation. This document is a source-based summary, not a replacement for the standard, project specifications, purchaser datasheet, material requisition, purchase order, approved deviations, or responsible engineering approval.

Prepared for engineering application and technical bid review

Document Control and Engineering Context

Document title	Centrifugal Pumps for Petroleum, Petrochemical, and Natural Gas Industries
Document number	API Standard 610
Edition / revision	Thirteenth Edition
Publication date	June 2026
Discipline	Mechanical / Rotating Equipment
Equipment	Centrifugal pumps: OH, BB, and VS types
Source file	API STD 610 13TH ED (2026).pdf
Decision supported	Pump selection, design review, technical bid evaluation, and vendor document review

Document hierarchy: API 610 Clause 5.3 requires the hierarchy of requirements to be specified. Where a Section 9 pump-type requirement conflicts with another section, Section 9 governs.

How to Use This Guide

- Identify the proposed API pump type and applicable Section 9 provisions.
- Enter all normal, rated, minimum, maximum, start-up, upset, cleaning, seasonal, and temporary-fluid cases.
- Apply every unconditional “shall” requirement and separately identify each purchaser decision marked “if specified.”
- Compare the vendor offer with the project document hierarchy and approved deviation list.
- Record objective evidence: curves, calculations, drawings, test reports, material certificates, NDE records, and purchaser approvals.
- Do not declare compliance where project inputs or vendor evidence are incomplete.

Key Numeric Constraints at a Glance

Design area	Constraint	Clause
Hydraulic selection	Rated flow: 80% to 110% of BEP; POR: 70% to 120% of BEP	6.1.22, 6.1.23
Head curve	Continuously rising to shutoff; minimum 10% rise	6.1.21
Future head	Minimum 5% head increase capability at rated flow	6.1.8
NPSH margin	Minimum 1 m at rated point unless otherwise specified	6.1.15
Variable speed	MCS at least 105% of rated speed; trip speed without damage	6.1.9, 6.1.10
Radially split casing	Triggered by ≥ 200 C, SG < 0.7 , > 100 bar(g), or > 55 C in < 10 s	6.2.16
Casing allowance	Minimum 3 mm corrosion allowance	6.2.14
Shaft deflection	Maximum 50 micrometres at primary seal faces	6.6.16
Dry critical speed	One/two-stage horizontal: $\geq 20\%$ above MCS	6.9.1.2
Vibration	OH/BB 3.0 mm/s RMS; VS 5.0 mm/s RMS within POR	Tables 8 and 9
Bearing system life	25,000 h rated; 16,000 h maximum loads	6.10.1.13

Design area	Constraint	Clause
Hydrotest	Minimum 1.5 x MAWP; minimum 30 min; no leakage/seepage	8.3.2
Performance tolerance	Rated head +/-3%; rated power +4%	Table 16

Detailed Design Constraints and Application Scenarios

1. API pump type and construction

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 4.2; 6.3; Table 4; Section 9.

Source requirement: The pump shall be classified by API type. OH1, OH4, OH5 and BB4 have special design features that prevent full compliance with specified general requirements and may be offered only if specified by the purchaser and supported by proven experience for the application.

Scenario: For a hot refinery end-suction service, an OH2 is proposed. If the vendor substitutes an OH1, the purchaser must have specified that type and the vendor must demonstrate the required experience.

Engineering considerations: The exact pump designation determines the applicable geometry, casing support, rotor, bearing, testing, and maintenance requirements. Section 9 governs where it conflicts with general clauses.

Verification required: Confirm the API type on the datasheet, GA and sectional drawing; review Table 4 limitations and the vendor deviation list.

2. Coverage of all operating cases

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.3; 6.1.7.

Source requirement: The equipment shall operate at the normal and rated points and at every other purchaser-specified operating condition. The purchaser shall specify operating conditions, liquid properties, preferred materials, site conditions, and utilities.

Scenario: A pump has normal, rated, start-up, minimum-recycle and maximum-flow cases. Suitability cannot be evaluated only at rated flow; all stated cases must be evaluated against hydraulic and mechanical limits.

Engineering considerations: Minimum flow can be limited by stability, vibration or temperature rise. Maximum flow can be limited by power, NPSH, AOR, vibration or driver capability.

Verification required: Plot every stated case on the vendor curve and record POR, AOR, MCSF, minimum thermal flow, NPSH3 and power.

3. Rated flow and preferred operating region

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.22; 6.1.23.

Source requirement: POR shall extend from 70% to 120% of BEP flow at rated impeller trim. Rated flow shall be between 80% and 110% of BEP flow.

Scenario: If BEP is 500 m³/h and rated flow is 380 m³/h, the rated point is 76% of BEP and fails the 80% lower limit even though it lies within the wider POR.

Engineering considerations: The rated-point criterion and POR criterion are separate checks. BEP must correspond to the rated impeller trim.

Verification required: Verify rated-flow/BEP ratio and annotate normal, rated, minimum and maximum flows on the vendor curve.

4. Continuously rising head curve

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.20; 6.1.21; 8.3.3.4.4.

Source requirement: The head curve shall rise continuously from rated point to shutoff with a minimum rise of 10%. A discharge orifice or device used to create a rising curve shall be stated in the proposal and requires purchaser approval.

Scenario: A pump with rated head 100 m and shutoff head 107 m has only 7% rise and does not meet the 10% requirement.

Engineering considerations: For pumps that cannot safely be tested at shutoff, the standard applies the rise requirement at the defined low-flow basis in Clause 8.3.3.4.4.

Verification required: Review proposal and certified curves, shutoff data, any curve-shaping device, and purchaser approval.

5. Future head capability

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.8.

Source requirement: The pump shall be capable of at least 5% head increase at rated flow by larger or different hydraulics, variable speed, or use of a blank stage.

Scenario: A pump already fitted with its maximum impeller and without speed margin must demonstrate another permitted means of achieving 5% additional head.

Engineering considerations: Additional head can affect pressure, driver power, NPSH, seal duty and system design pressure; these interfaces require separate verification.

Verification required: Obtain minimum/rated/maximum impeller curves or speed/stage basis and repeat pressure and power checks at the future condition.

6. NPSH₃, datum and minimum margin

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.12 to 6.1.16.

Source requirement: Vendor NPSH₃ shall be based on water below 55 C without correction for hydrocarbons. At rated point, NPSHA minus NPSH₃ shall be at least 1 m unless the purchaser specifies another margin. Applicable NPSH datum elevations shall be used.

Scenario: NPSHA 6.0 m and NPSH₃ 5.4 m provide only 0.6 m margin and do not meet the baseline 1 m requirement. A datum mismatch can also invalidate the comparison.

Engineering considerations: The 1 m value is a minimum; the purchaser may specify a different margin. Vertical pump datum depends on pump type.

Verification required: Check process NPSHA calculation, vendor NPSH₃ curve, common datum, rated speed/flow and optional NPSH test requirement.

7. Parallel operation

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.18.

Source requirement: When parallel operation is specified, head values at a given flow within POR shall be within 3% of each other for pumps with discharge larger than DN 80.

Scenario: Two applicable pumps differ by 5% head at the same flow within the intended parallel range; they do not satisfy the 3% matching requirement.

Engineering considerations: Matching only the rated head is insufficient; comparison is required across the common operating range within POR.

Verification required: Overlay certified curves at equal speed and impeller condition; document the head difference over the intended parallel range.

8. Maximum continuous speed and trip speed

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.9; 6.1.10; 6.1.24.

Source requirement: Fixed-speed pumps shall meet the defined maximum supply-frequency basis. Variable-speed pumps and specified spare arrangements shall have MCS at least 105% of rated speed. Variable-speed pumps shall reach trip speed without damage. Vendor shall state ASD minimum and maximum allowable speeds.

Scenario: A 3000 rpm ASD pump must be evaluated for the defined MCS and trip speed, not only at 3000 rpm.

Engineering considerations: Speed affects head, pressure, power, rotor dynamics, bearing operation, seal duty and vibration.

Verification required: Record minimum speed, rated speed, MCS and trip speed; review curves, pressure calculation and dynamic analyses across the range.

9. Maximum discharge pressure and MAWP

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.2.1 to 6.2.13.

Source requirement: Vendor shall calculate maximum discharge pressure for all operating cases using maximum differential pressure, applicable density, maximum suction pressure and limiting speed. Except as provided, **MAWP shall be at least maximum discharge pressure plus 10% of maximum differential pressure and shall meet the applicable minimum rating.**

Scenario: An upset combines maximum suction pressure, maximum speed and a dense temporary liquid. The casing cannot be rated only for the normal case.

Engineering considerations: Temporary higher-density liquid and maximum impeller/stages are included when specified. Seal chamber and gland require pressure-temperature rating at least equal to the attached casing.

Verification required: Obtain a case-by-case pressure summary showing suction pressure, differential pressure, density, speed, impeller/stages, discharge pressure, MAWP, temperature basis and flange rating.

10. System design pressure interface

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.19.

Source requirement: If the purchaser specifies system design pressure, pump maximum discharge pressure shall not exceed it.

Scenario: A pump can develop 38 bar(g) while a stated downstream system design pressure is 35 bar(g). The mismatch must be resolved through the project authority; the source does not authorize ignoring either value.

Engineering considerations: The pump calculation must include the applicable operating cases before comparison with the system pressure.

Verification required: Compare the vendor maximum-discharge-pressure calculation with stated upstream and downstream design pressures.

11. Radially split casing triggers

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.2.16.

Source requirement: Unless otherwise specified, radially split casings shall be supplied for any of these:

- pumping temperature 200 C or higher;
- liquid relative density below 0.7;
- rated discharge pressure above 100 bar(g); or
- temperature transient greater than 55 C in less than 10 seconds.

Scenario: A hydrocarbon pump at 220 C, an LPG pump with SG 0.58, a pump above 100 bar(g), or a standby pump exposed to a 70 C change in 8 seconds each triggers this requirement.

Engineering considerations: The note regarding successful axially split applications does not remove the “shall” requirement. Any departure must follow the project hierarchy and approval process.

Verification required: Check maximum temperature, minimum SG, rated discharge pressure and fastest specified thermal transient against the proposed casing split.

12. Pressure casing structural basis

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.2.4 to 6.2.8; 6.2.14.

Source requirement: The casing shall operate without leakage or internal contact at MAWP, maximum operating temperature and the worst combination of twice Table 5 nozzle loads. Minimum casing corrosion allowance is 3 mm unless an accepted alternative material basis applies.

Scenario: Piping loads meet 1.0 times Table 5. The casing still requires a 2.0 times Table 5 structural design basis; the higher value is not the normal piping allowable.

Engineering considerations: Allowable stress and casting factor depend on the selected design method and NDE extent.

Verification required: Review casing calculation, material properties, casting factor, corrosion allowance, nozzle-load basis and hydrotest report.

13. External nozzle forces and moments

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.5; Table 5; Annex F.

Source requirement: Applicable pumps and baseplates shall operate with Table 5 loads applied simultaneously at suction and discharge in the worst combination. Annex F may qualify loads beyond Table 5, requires purchaser approval, and does not remove the 2x casing margin.

Scenario: A piping model exceeds one Table 5 moment. The load must be reduced or evaluated under all Annex F interaction and translated-load criteria with the required approval.

Engineering considerations: “Two times API load” shall not be used as a blanket piping allowable. Coordinate system and nozzle orientation are essential.

Verification required: Check each force/moment, resultants, pump-center translations where used, coordinate system, vendor confirmation and purchaser approval.

14. Nozzle and auxiliary connections

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.4.1 to 6.4.3.

Source requirement: The specified nonstandard nozzle sizes shall not be used. Suction and discharge flanges normally have equal ratings. Flange faces shall be parallel to the GA plane within 0.5 degrees. Threaded pressure-casing connections are generally prohibited except stated cases; vents and drains are required unless self-venting.

Scenario: A sour-service pump is offered with socket-welded casing connections. NACE service is one of the triggers for full-penetration butt welds in Clause 6.4.3.3.

Engineering considerations: Connection method, rating, material, accessibility, gusseting, venting, draining and NDE are linked requirements.

Verification required: Review nozzle schedule, connection table, flange rating, weld detail, NDE, vent and drain arrangement.

15. Shaft construction and runout

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.6.9; 6.6.10; 9.3.3.

Source requirement: Horizontal pump shafts shall normally be single-piece with TIR no greater than 25 micrometres. VS shaft TIR shall not exceed 40 micrometres per metre in sections 1.2 m or less; multi-piece construction requires purchaser approval where permitted.

Scenario: A long VS shaft is split for shipping. Purchaser approval is required, and the applicable VS runout records remain necessary.

Engineering considerations: Runout limits differ between general horizontal and VS provisions; apply the pump-type rule.

Verification required: Review shaft drawing, runout report, shaft joints, coupling details and purchaser approval.

16. Shaft deflection at seal faces

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.6.16.

Source requirement: Shaft stiffness shall limit total deflection at primary seal faces to 50 micrometres under the most severe dynamic condition over AOR, using maximum-diameter impeller, specified speed and fluid conditions.

Scenario: A long-overhang OH2 is offered with a small rated impeller. The deflection check must still use the maximum-diameter impeller required by the clause.

Engineering considerations: Shaft diameter, span, overhang and hydraulic casing design can influence compliance.

Verification required: Obtain the deflection calculation and verify impeller size, speed, liquid density, operating range and force basis.

17. Critical-speed separation

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.9.1.2; 9.3.12.2.

Source requirement: For one- and two-stage horizontal pumps, first dry bending critical speed shall be at least 20% above MCS unless Annex I analysis is used. VS5 first dry critical speed shall be 30% above maximum allowable continuous speed.

Scenario: A horizontal pump with MCS 3600 rpm and dry critical speed 4100 rpm has less than the required 20% separation and needs redesign or the permitted Annex I route.

Engineering considerations: Use maximum continuous speed, not only rated speed.

Verification required: Review dry critical speed, MCS, pump type, lateral-analysis trigger and alternative assessment evidence.

18. Lateral rotor analysis

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 9.2.4.1; Annex I.

Source requirement: For applicable BB pumps, analysis is unnecessary only when the pump/service is identical to proven equipment or the rotor is subcritical. Otherwise analysis is required. Where damping/separation is inadequate, unbalance response is required and peak-to-peak displacement shall not exceed 35% of diametral running clearance.

Scenario: A new BB5 rotor is neither identical to proven equipment nor shown to be subcritical; lateral analysis is therefore required.

Engineering considerations: The model includes specified clearance states, pumped liquid effects, bearing/support properties and coupling mass effects.

Verification required: Review rotor model, coefficients, Campbell diagram, damping plot, mode shapes, unbalance response and clearance-displacement comparison.

19. Component and rotor balance

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.9.3; 9.2.4.2; Annex J.

Source requirement: Major rotating components shall be dynamically balanced to ISO 21940-11 Grade G2.5. Multistage BB rotors and one/two-stage BB rotors above 3800 rpm require two-plane assembled-rotor balancing as specified. Assembly balancing requires residual-unbalance verification.

Scenario: A five-stage BB5 cannot be accepted on component certificates alone; the applicable assembled-rotor balance and residual-unbalance records are required.

Engineering considerations: Balance grade and procedure vary with component fit, speed and rotor flexibility.

Verification required: Obtain component certificates, rotor balance report, half-key record, residual-unbalance worksheet and final runout.

20. Wear rings and running clearances

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.7; Table 6.

Source requirement: Renewable casing wear rings shall be provided. Metallic mating surfaces require at least 50 HB difference unless both are at least 400 HB. Table 6 minimum diametral clearances apply; add 0.125 mm for higher-galling materials or operation above 260 C.

Scenario: For a 120 mm rotating diameter, Table 6 gives 0.40 mm minimum diametral clearance. A high-galling or above-260 C case requires the stated 0.125 mm addition.

Engineering considerations: Smaller nonmetallic clearances may be proposed only on the documented basis and purchaser approval described by the standard.

Verification required: Check as-built diameter, clearance, materials, hardness, temperature, galling tendency and approved exceptions.

21. Mechanical seal and chamber

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.8; API 682 interface.

Source requirement: Pumps shall use API 682 seals. Seal interfaces shall meet Tables 7A/7B as applicable; cartridge removal shall generally not require moving the driver. Register concentricity shall be ≤ 125 micrometres TIR and chamber face runout ≤ 0.5 micrometres/mm of bore; complete venting is required.

Scenario: An OH2 requires motor removal to withdraw the seal cartridge. This conflicts with the general cartridge-removal requirement.

Engineering considerations: Static and dynamic seal-chamber pressures must be determined and agreed by purchaser and vendor.

Verification required: Review seal drawing, API 682 category/plan, chamber dimensions, pressure calculation, runouts, venting and extraction clearance.

22. Vibration limits and operating region

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.9.4; Tables 8 and 9.

Source requirement: Within POR, OH/BB pumps within the Table 8 stated range shall remain below 3.0 mm/s RMS; VS pumps below 5.0 mm/s RMS. Outside POR but within AOR, a 30% increase is permitted. Pumps with proximity probes shall meet housing and shaft limits.

Scenario: An OH2 records 3.5 mm/s RMS at a point claimed inside POR. The 30% allowance cannot be used because it applies only outside POR within AOR.

Engineering considerations: Acceptance requires the specified test locations, overall values, discrete frequencies and FFT spectra.

Verification required: Check all required test points at POR/AOR boundaries and rated flow, measurement locations, spectra and instrument basis.

23. High-energy pump provisions

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 3.1.12; 6.1.26; Annex D.

Source requirement: A high-energy pump has head above 200 m per stage and power above 225 kW per stage. Minimum impeller-to-collector radial clearance is 3% of vane-tip radius for diffusers and 6% for volute cutwaters.

Scenario: A stage with 240 m head and 260 kW meets both thresholds and triggers the high-energy provisions.

Engineering considerations: Annex D provides additional special-purpose guidance; it is informative unless invoked by the purchaser or project documents.

Verification required: Check head/stage, power/stage, vane-tip radius, collector clearance, vibration provisions and specified special-purpose reviews.

24. Bearing arrangement and energy density

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.10.1; Table 10.

Source requirement: Each shaft shall generally have two radial bearings and one double-acting thrust bearing. Non-pipeline pumps at or above 4.0×10^6 kW-rpm energy density require hydrodynamic radial and thrust bearings. Pipeline thresholds and mixed arrangements are separately defined.

Scenario: A non-pipeline process pump exceeds the energy-density threshold; a fully rolling-element arrangement is not the baseline compliant selection.

Engineering considerations: Bearing manufacturer speed limits and pump-type provisions also apply.

Verification required: Check rated pump power, speed, pipeline classification, bearing type, bearing speed limit, thrust loads and approvals.

25. Bearing system life

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.10.1.12 to 6.10.1.15.

Source requirement: Rolling-element bearing system life shall be at least 25,000 h at rated conditions and 16,000 h at maximum radial/axial loads and rated speed.

Scenario: The vendor quotes 25,000 h for each individual bearing but does not calculate combined system life. This is insufficient evidence for the system-life requirement.

Engineering considerations: Loads include design and twice-design clearances, coupling thrust, differential pressure and applicable train reactions.

Verification required: Obtain system-life calculation, individual lives, loads, clearance cases, speed and bearing data.

26. Bearing temperatures and housing sealing

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.10.2.7 to 6.10.2.9.

Source requirement: Pressurized-oil temperature rise shall be ≤ 28 K; bearing metal temperature, where instrumented, ≤ 93 C; non-pressure/oil-mist housing surface ≤ 70 C. Rolling-element housings require replaceable labyrinth or magnetic end seals; lip seals are prohibited.

Scenario: A splash-lubricated housing stabilizes at 76 C during testing, above the 70 C limit.

Engineering considerations: Solar radiation is noted as a potential contributor in field service, but the source limit remains the stated requirement.

Verification required: Review temperature trend, ambient basis, lubrication method, cooling, housing seal type and test report.

27. Cooling-water design basis

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.31; Table 2.

Source requirement: Water-side design conditions include 1.5 to 2.5 m/s velocity, minimum 7 bar(g) MAWP, 10.5 bar(g) test pressure, maximum 1 bar pressure drop, 30 C inlet, 50 C outlet and 20 C rise.

Scenario: Project cooling water arrives at 35 C, above the 30 C table basis. Vendor and purchaser must address the actual utility condition and required flow.

Engineering considerations: Cooling circuits require complete venting and draining; parallel cooler arrangement is required by the piping provisions.

Verification required: Compare project utilities with Table 2 and review thermal duty, flow, pressure drop, materials, fouling, venting, draining and test pressure.

28. Motor rating margin

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 7.1.1 to 7.1.9; Table 12.

Source requirement: Motor nameplate rating, excluding service factor, shall be rated pump power times 1.25 below 22 kW, 1.15 from 22 to 55 kW, and 1.10 above 55 kW, then rounded to the next standard size. Minimum motor rating is 4 kW unless an alternative is approved.

Scenario: A pump requiring 50 kW at rated condition has a Table 12 basis of 57.5 kW before rounding; another specified condition may still govern.

Engineering considerations: Driver sizing includes specified process variations and applicable seal, bearing, gear and coupling losses.

Verification required: Check rated and maximum curve power, liquid density, future impeller, speed range, losses, start condition and motor standard size.

29. Motor acceleration

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 7.1.11.

Source requirement: The motor shall accelerate the pump to rated speed at 80% voltage against a closed discharge valve.

Scenario: An electrical study predicts reduced terminal voltage. Vendor must demonstrate acceleration at the API condition, not only at nominal voltage.

Engineering considerations: The demonstration depends on motor torque-speed data, pump load torque, inertia and starting method.

Verification required: Review motor speed-torque curve, pump torque-speed curve, inertia and acceleration calculation at 80% voltage.

30. Coupling design and maintenance space

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 7.2.

Source requirement: Couplings shall generally be all-metal flexible-disc spacer type to AGMA 9000 Class 9 with minimum service factor 1.5. OH pump DBSE shall be at least 125 mm and long enough for the longest removable component. Above 3800 rpm, API 671 balance provisions apply.

Scenario: An OH2 has 100 mm DBSE and a 115 mm seal cartridge; it fails both the minimum distance and maintenance-removal intent.

Engineering considerations: Coupling rating is based on driver nameplate power. Fit, keys and balance requirements also apply.

Verification required: Check DBSE, seal removal envelope, coupling power rating, service factor, speed, fit, keys and guard.

31. Baseplate stiffness and alignment

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 7.4.25; 7.4.26; Table 13.

Source requirement: Baseplate stiffness shall limit pump-shaft displacement under specified nozzle moments. Grouted base acceptance is 175 micrometres in +Z and 75 micrometres in -Y; non-grouted values are 125 and 50 micrometres. Grout shall not be used to obtain stiffness during the test.

Scenario: A vendor FEA meets displacement only after including grout stiffness. That does not satisfy the stated test basis.

Engineering considerations: Physical testing or FEA is performed when specified. Support and anchor conditions must match the required basis.

Verification required: Review baseplate type, support points, applied moments, absolute shaft displacement and specified test/FEA evidence.

32. Mounting surfaces and soft-foot control

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.2.26; 7.4.12 to 7.4.15.

Source requirement: Equipment mounting surfaces require 6 micrometres Ra or better, 75 micrometres/m flatness, and stated coplanarity/parallelism limits. Pumps are mounted directly on baseplate pads without shims; drive-train components receive shims or specified chocks.

Scenario: A vendor proposes shims under pump feet to correct an unlevel baseplate. The standard states that shims shall not be used under the pump.

Engineering considerations: The baseplate flatness demonstration is made in the vendor shop under the specified support condition.

Verification required: Review pad inspection, baseplate flatness, pump foot contact, shim arrangement and alignment report.

33. Maintenance and rotor withdrawal

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.1.40; pump-type exceptions in Section 9.

Source requirement: Except for vertically suspended and integrally geared pumps, the rotor or inner bundle shall be removable without disconnecting suction/discharge piping or moving the driver.

Scenario: A BB pump arrangement requires moving the driver to withdraw the bundle; this conflicts with the general maintainability requirement unless an applicable specific provision or approved deviation governs.

Engineering considerations: Maintenance clearance, lifting and spool-piece arrangements must be shown on vendor drawings.

Verification required: Review GA, withdrawal route, DBSE, lifting method, removable spool needs, clearance and heaviest maintenance lift.

34. Material class and compatibility

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.12; Annexes G and H.

Source requirement: Purchaser shall specify an Annex H material class. Vendor shall identify exact component specifications and consider corrosion, erosion, stress-corrosion cracking, elastomers, galvanic effects and service temperature.

Scenario: For seawater, Annex G lists possible classes but requires purchaser-vendor agreement on materials best suited to intended use; the guidance table alone does not complete selection.

Engineering considerations: Annex G is informative guidance; Annex H material requirements are normative when the class is selected.

Verification required: Review fluid analysis, service temperature, solids, material class, exact grades, heat treatment, hardness, PMI, NDE and welding.

35. Wet H₂S and reduced-hardness materials

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.12.1.14 to 6.12.1.16.

Source requirement: Purchaser shall state wet H₂S exposure. When specified, NACE MR0103 or MR0175 applies according to the service basis. Ferrous materials outside their coverage shall not exceed 620 MPa yield strength and HRC 22.

Scenario: A refinery sour-water pump is described only as “NACE compliant.” The governing NACE document and exposure basis must be stated.

Engineering considerations: Welds and heat-affected zones may require PWHT to meet hardness and strength requirements.

Verification required: Confirm facility/service basis, H₂S, chlorides as relevant, NACE document, materials, hardness map, WPS and PWHT.

36. Minimum design metal temperature

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 6.12.4.

Source requirement: Purchaser shall specify MDMT and select the governing impact-test basis. Materials shall be suitable for the MDMT, considering ambient, liquid and possible auto-refrigeration.

Scenario: A liquefied-hydrocarbon pump operates above 0 °C but can auto-refrigerate to -45 °C during depressurization; the lower condition must be considered for MDMT.

Engineering considerations: Governing thickness and impact-test exemptions follow the selected code basis identified in the clause.

Verification required: Check minimum ambient/liquid temperatures, depressurization, MDMT, governing thickness, impact tests, bolting and lifting materials.

37. Hydrostatic test

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 8.3.2.

Source requirement: Pressure-casing assemblies shall be tested at a minimum of 1.5 times MAWP for at least 30 minutes with no leakage or seepage. Mechanical seals are excluded from casing hydrotest. Stainless-steel test liquid shall not exceed 50 ppm chloride.

Scenario: A 40 bar(g) MAWP casing tested at 55 bar(g) is below the unadjusted 60 bar(g) minimum.

Engineering considerations: Material-temperature stress correction can affect the required pressure as stated in Clause 8.3.2.7 and must be documented where applicable.

Verification required: Review MAWP, test pressure, duration, temperature/stress correction, chloride content, gasket, isolation and report.

38. Performance-test acceptance

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 8.3.3; Table 16.

Source requirement: Each pump shall receive a performance test to the stated test code. Rated differential head tolerance is $\pm 3\%$, rated power tolerance is $+4\%$, and rated NPSH3 shall not exceed the quoted value. Test speed shall be within 3% of rated and corrected to rated speed.

Scenario: A tested rated head 4% below guarantee is outside the $\pm 3\%$ tolerance. Power 3% above quoted is within the $+4\%$ limit, subject to all other acceptance requirements.

Engineering considerations: Acceptance also includes vibration, temperatures, leakage and required curve shape.

Verification required: Review raw log, fitted curve, correction equation, speed, water temperature, head, power, vibration, seal leakage and guarantees.

39. NPSH test when specified

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 8.3.5.3.

Source requirement: When specified, NPSH3 is determined at required points from 3% head drop, or 3% first-stage head drop for multistage pumps. Test shall not proceed beyond 20% breakdown. Rated NPSH3 shall not exceed quoted value.

Scenario: For a multistage pump, dividing total head by stage count to estimate first-stage head requires purchaser approval.

Engineering considerations: Direct first-stage measurement or first-stage-only testing is preferred by the stated provisions where possible.

Verification required: Review test method, points, starting NPSHA, head-vs-NPSHA curves, first-stage basis, approval and final NPSH3 curve.

40. Post-test modification and final integrity

Applicable source: API Standard 610, 13th Edition (June 2026), Clause 8.3.3.12; 8.3.3.13.

Source requirement: Impeller diameter reduction above 5% after test requires retest. Vane machining after performance test requires purchaser approval and full retest. Other corrective dismantling invalidates the initial test and requires final retest. Pumps shall not be disassembled after final test except stated provisions.

Scenario: Vendor underfiles impeller vanes after witnessed test and proposes shipment without retest. The standard requires purchaser approval, rebalance as applicable, and full retest.

Engineering considerations: Disturbing the seal after test triggers the specified final seal air test.

Verification required: Review post-test work, final impeller dimensions, machining method, rebalance, retest, seal air test and preservation records.

Purchaser-Specified Decisions to Capture

API 610 uses a bullet symbol to identify provisions requiring purchaser information or a purchaser decision. The following project inputs should be captured in the requisition and datasheet where applicable. This list is a focused review aid, not a substitute for checking every bullet-marked provision in the standard.

- Operating cases, liquid properties, site/environmental conditions and utilities.
- NPSHA, any NPSH margin different from 1 m, and any suction-specific-speed limit.
- Parallel operation, system design pressure, ASD speed range and maximum allowable sound pressure level.
- Pressure cases involving temporary higher-density liquid, maximum impeller/stages or dual pressure rating.
- Seal category, seal system, monitoring and documentation requirements.
- Nozzle connection construction, auxiliary connection details and optional Annex F qualification.
- Material class, MDMT, impact-test basis, H₂S conditions, NACE basis, certificates, PMI and optional NDE.
- Bearing instrumentation, vibration probes, temperature sensors, lubrication method and auxiliary systems.
- Baseplate type, optional stiffness test or FEA, steel-module mounting and alignment requirements.
- Inspection level, hold/witness/observe points, optional mechanical run, NPSH, complete-unit, sound, resonance and spare-part tests.
- Vendor document requirements, proposal deviations, analyses and final test reports.

Project Compliance Matrix Template

No.	API clause	Requirement / check	Project input	Vendor evidence	Status / action
1	4.2 / 9	API pump type and applicable pump-type clauses			
2	6.1.7	All operating cases within declared limits			
3	6.1.22-23	Rated flow/BEP and POR			
4	6.1.21	Rising curve and 10% rise			
5	6.1.8	5% future head capability			
6	6.1.12-15	NPSHA, NPSH ₃ , datum and margin			
7	6.2.1-14	Maximum pressure, MAWP and corrosion allowance			
8	6.2.16	Casing split triggers			
9	6.5 / F	Nozzle loads and approvals			
10	6.6-6.9	Shaft, clearances, balance and dynamics			
11	6.8 / API 682	Seal chamber and seal system			

No.	API clause	Requirement / check	Project input	Vendor evidence	Status / action
12	6.10-6.11	Bearings, life, temperatures and lubrication			
13	6.12 / H	Materials, MDMT, NACE and NDE			
14	7.1	Driver sizing and acceleration			
15	7.2-7.4	Coupling, baseplate and maintenance			
16	8.3	Hydrotest, performance and optional tests			

How to Select a Centrifugal Pump (API 610)

Selection principle

Do not select a pump only because one vendor point meets rated flow and head. The complete selection must satisfy service suitability, BEP location, operating range, NPSH, power, pressure rating, materials, sealing, auxiliaries and testing.

Revision control note: Confirm the project-approved revision and standards hierarchy before applying this guide to an active project.

1. Selection Workflow

1. Confirm applicability and document hierarchy.
2. Collect complete process, liquid and site data.
3. Define minimum, lowest-normal, dominant-normal, rated and maximum flow cases.
4. Calculate total differential head for every relevant operating case.
5. Develop or understand the system curve and classify the system as static-head or friction dominated.
6. Calculate NPSHA at the worst suction condition.
7. Select the service-appropriate pump configuration and specific restrictions.
8. Review BEP location, POR/AOR coverage, MCSF and minimum thermal flow.
9. Verify NPSH3, required NPSH margin and suction-specific speed.
10. Calculate absorbed power and select the driver against the full operating envelope.
11. Verify pressure casing, temperature, materials, seal arrangement and auxiliaries.
12. Define inspection and test requirements, then complete the vendor-curve acceptance review.

Recommended review sequence

Process basis -> Hydraulic calculation -> Pump type -> Curve check -> NPSH -> Power -> Mechanical design -> Materials and seals -> Auxiliaries -> Testing -> Deviations/TQs

2. Minimum Data Required Before Selection

Data group	Required information
Operating cases	Minimum, lowest-normal, dominant-normal, rated, maximum, start-up, upset, flushing and standby conditions.
Hydraulic data	Flow, suction pressure, discharge pressure, static level, elevation, equipment pressure drops and control-valve allocation.
Liquid data	Density, viscosity, vapor pressure, temperature, solids, gas content, chlorides, H2S, toxicity, flammability and corrosivity.
Site data	Onshore/offshore/coastal location, ambient temperature, installation elevation, hazardous-area classification and electrical supply.
Reliability data	Continuous/intermittent duty, operating hours, standby philosophy, parallel operation and required service life.

3. Define Normal and Rated Flow

3.1 Dominant normal flow

Use the PFD heat-and-material balance condition at which the pump will operate for most of its service life. When more than one dominant case exists, retain all significant normal cases for vendor-curve review rather than choosing a single convenient point.

3.2 Rated flow margin

For a normal process pump:

$$Q_r \leq 1.10 Q_n$$

For a recycle or circulation pump, a margin up to 20% may be used when justified:

$$Q_r \leq 1.20 Q_n$$

where:

- Q_r = rated flow, m³/h
- Q_n = dominant normal flow, m³/h

Avoid oversize

A short-duration high-flow case should normally be identified as a mandatory operating point rather than automatically becoming the rated point. Excess rated-flow margin can lead to oversized pumps, throttling loss, poor BEP location and higher NPSH demand.

Example: rated-flow margin

Dominant normal flow = 200 m³/h. For a normal process pump:

$$Q_{\text{rated, max}} = 1.10 \times 200 = 220 \text{ m}^3/\text{h}$$

A rated point of 220 m³/h is acceptable. A proposed 250 m³/h rated point requires an explicit process basis rather than an arbitrary future margin.

4. Calculate Total Differential Head

- For points at pump suction and discharge:

$$H_p = \frac{P_d - P_s}{\rho g} + \frac{V_d^2 - V_s^2}{2g} + (z_d - z_s)$$

- For the complete piping system:

$$H_{\text{required}} = H_{\text{static}} + H_{\text{pressure}} + H_{\text{friction}} + H_{\text{equipment}} + H_{\text{control valve}}$$

- Calculate head for minimum and maximum source liquid levels.
- Check clean and specified fouled piping or equipment conditions.
- Evaluate minimum, normal, rated and maximum flow cases.
- Include start-up, flushing, parallel operation and abnormal header pressures where applicable.
- Do not add a second arbitrary head margin when process pressure drops already contain conservatism.

Example:**Given data**

Parameter	Value
Rated flow, Q	200 m ³ /h
Liquid	Water
Density, ρ	1000 kg/m ³
Suction vessel pressure, P_1	0.5 bar(g)
Discharge vessel pressure, P_2	3.0 bar(g)
Suction liquid-level elevation, z_1	5 m
Discharge liquid-level elevation, z_2	25 m
Suction-piping loss	2 m
Discharge-piping loss	12 m
Heat-exchanger loss	8 m
Control-valve loss	5 m
Gravitational acceleration, g	9.81 m/s ²

Because both calculation points are taken at large vessel liquid surfaces, the surface velocities are approximately zero:

$$V_1 \approx V_2 \approx 0$$

Therefore, the velocity-head difference can initially be neglected.

TDH equation

For transfer between two vessels:

$$TDH = \frac{P_2 - P_1}{\rho g} + (z_2 - z_1) + h_{f,s} + h_{f,d} + h_{\text{equipment}} + h_{\text{control valve}}$$

Where:

- TDH = total differential head, m
- P_1 = pressure acting on suction-vessel liquid surface, Pa
- P_2 = pressure acting on discharge-vessel liquid surface, Pa
- ρ = liquid density, kg/m³
- g = gravitational acceleration, m/s²
- z_1 = suction-vessel liquid-level elevation, m
- z_2 = discharge-vessel liquid-level elevation, m
- $h_{f,s}$ = suction-piping friction loss, m
- $h_{f,d}$ = discharge-piping friction loss, m
- $h_{\text{equipment}}$ = equipment pressure-drop head, m
- $h_{\text{control valve}}$ = control-valve pressure-drop head, m

Step 1: Calculate pressure head

The pressure difference between the receiving and suction vessels is:

$$\begin{aligned}\Delta P &= P_2 - P_1 \\ \Delta P &= 3.0 - 0.5 \\ \Delta P &= 2.5 \text{ bar}\end{aligned}$$

Convert bar to pascals:

$$\begin{aligned}1 \text{ bar} &= 100000 \text{ Pa} \\ \Delta P &= 2.5 \times 100000 \\ \Delta P &= 250000 \text{ Pa}\end{aligned}$$

Pressure head:

$$\begin{aligned}H_{\text{pressure}} &= \frac{\Delta P}{\rho g} \\ H_{\text{pressure}} &= \frac{250000}{1000(9.81)} \\ H_{\text{pressure}} &= 25.48 \text{ m}\end{aligned}$$

Shortcut for water

For water:

$$1 \text{ bar} \approx 10.2 \text{ mH}_2\text{O}$$

Therefore:

$$\begin{aligned}H_{\text{pressure}} &\approx 2.5(10.2) \\ H_{\text{pressure}} &\approx 25.5 \text{ m}\end{aligned}$$

5. Step 2: Calculate static elevation head

$$\begin{aligned}H_{\text{elevation}} &= z_2 - z_1 \\ H_{\text{elevation}} &= 25 - 5 \\ H_{\text{elevation}} &= 20 \text{ m}\end{aligned}$$

Notice that we use the difference between the two liquid levels, not the discharge-vessel elevation measured from ground level alone.

Step 3: Calculate piping-friction head

The suction-piping loss is:

$$h_{f,s} = 2 \text{ m}$$

The discharge-piping loss is:

$$h_{f,d} = 12 \text{ m}$$

Therefore:

$$\begin{aligned}H_{\text{piping}} &= h_{f,s} + h_{f,d} \\ H_{\text{piping}} &= 2 + 12 \\ H_{\text{piping}} &= 14 \text{ m}\end{aligned}$$

Step 4: Add heat-exchanger loss

$$H_{\text{HX}} = 8 \text{ m}$$

This may originally be supplied by the process engineer as a pressure drop.

For example, if the heat-exchanger pressure drop were given as bar:

$$\Delta P_{HX} = 0.785 \text{ bar}$$

For water:

$$H_{HX} = \frac{0.785 \times 100000}{1000(9.81)}$$
$$H_{HX} \approx 8.0 \text{ m}$$

Step 5: Add control-valve head

$$H_{CV} = 5 \text{ m}$$

The control-valve pressure drop should be provided or confirmed by the process or control-valve engineer. It must not be confused with piping friction.

For water, 5 m corresponds approximately to:

$$\Delta P_{CV} = \rho g H$$
$$\Delta P_{CV} = 1000(9.81)(5)$$
$$\Delta P_{CV} = 49050 \text{ Pa}$$
$$\Delta P_{CV} \approx 0.49 \text{ bar}$$

Step 6: Calculate total differential head

Now add all components:

$$TDH = H_{\text{pressure}} + H_{\text{elevation}} + H_{\text{piping}} + H_{HX} + H_{CV}$$
$$TDH = 25.48 + 20 + 14 + 8 + 5$$
$$TDH = 72.48 \text{ m}$$

Rounded for the pump datasheet:

$$TDH \approx 72.5 \text{ m}$$

The required rated duty is therefore:

$$Q = 200 \text{ m}^3/\text{h at } H = 72.5 \text{ m}$$

Calculate approximate pump shaft power

Assume the vendor's pump efficiency at the rated point is:

$$\eta_p = 80\% = 0.80$$

For flow in m^3/h :

$$P_{\text{shaft}} = \frac{\rho Q H}{367000 \eta_p}$$

Substitute:

$$P_{\text{shaft}} = \frac{1000(200)(72.48)}{367000(0.80)}$$
$$P_{\text{shaft}} = 49.37 \text{ kW}$$
$$P_{\text{shaft}} \approx 49.4 \text{ kW}$$

This is the absorbed pump power at the stated rated point. It is **not yet the final motor rating**. The motor must also be checked against:

- Maximum specified flow
- End-of-curve power
- Maximum impeller diameter
- Maximum-density liquid
- Maximum speed, if variable speed
- Start-up viscosity
- Flushing or commissioning liquid

Cross-check using pump nozzle pressures

Suppose the selected pump has:

Suction nozzle pressure : 0.30 bar(g)

Discharge nozzle pressure: 7.20 bar(g)

Suction pipe internal diameter: 0.20 m

Discharge pipe internal diameter: 0.15 m

Discharge pressure tapping is 0.30 m above the suction tapping

For nozzle-based TDH, use:

$$H_p = \frac{P_d - P_s}{\rho g} + \frac{V_d^2 - V_s^2}{2g} + (z_d - z_s)$$

Flow conversion

$$Q = \frac{200}{3600}$$

$$Q = 0.05556 \text{ m}^3/\text{s}$$

Suction velocity

$$A_s = \frac{\pi(0.20)^2}{4}$$

$$A_s = 0.03142 \text{ m}^2$$

$$V_s = \frac{0.05556}{0.03142}$$

$$V_s = 1.77 \text{ m/s}$$

Discharge velocity

$$A_d = \frac{\pi(0.15)^2}{4}$$

$$A_d = 0.01767 \text{ m}^2$$

$$V_d = \frac{0.05556}{0.01767}$$

$$V_d = 3.14 \text{ m/s}$$

Pressure-head difference

$$P_d - P_s = 7.20 - 0.30 = 6.90 \text{ bar}$$

$$H_p = \frac{6.90 \times 100000}{1000(9.81)}$$

$$H_p = 70.34 \text{ m}$$

Velocity-head difference

$$\Delta H_v = \frac{V_d^2 - V_s^2}{2g}$$

$$\Delta H_V = \frac{3.14^2 - 1.77^2}{2(9.81)}$$

$$\Delta H_V = \frac{9.86 - 3.13}{19.62}$$

$$\Delta H_V = 0.34 \text{ m}$$

Elevation correction

$$z_d - z_s = 0.30 \text{ m}$$

Pump head

$$H_p = 70.34 + 0.34 + 0.30$$

$$\boxed{H_p = 70.98 \text{ m}}$$

This nozzle-based result is close to the system requirement of 72.5 m. Any difference should be reconciled through actual operating conditions, measurement locations, instrument accuracy and loss assumptions.

Important engineering rules

Rule 1: Use pressure difference, not discharge pressure alone

Incorrect:

$$H = \frac{P_d}{\rho g}$$

Correct:

$$H_{\text{pressure}} = \frac{P_d - P_s}{\rho g}$$

The pump receives credit for existing suction pressure.

Rule 2: Use liquid-level elevations for vessel-to-vessel calculations

For transfer between vessels:

$$H_{\text{static}} = z_2 - z_1$$

Do not calculate static head only from the pump centerline unless the selected energy-balance points require it.

Rule 3: Add losses only once

- Avoid double-counting:
- Piping loss included in the process hydraulic calculation
- Equipment pressure drop included separately
- Control-valve pressure drop included separately
- Then an additional unexplained “head margin”
- All assumptions should be visible and traceable.

Rule 4: Pressure head depends on liquid density

For the same pressure difference:

$$H = \frac{\Delta P}{\rho g}$$

If density decreases, equivalent head increases.

For example, for hydrocarbons with:

$$\rho = 800 \text{ kg/m}^3$$

and:

$$\Delta P = 2.5 \text{ bar}$$
$$H_{\text{pressure}} = \frac{250000}{800(9.81)}$$
$$H_{\text{pressure}} = 31.86 \text{ m}$$

The same 2.5 bar pressure difference equals:

25.48 m of water

31.86 m of hydrocarbon at 800 kg/m³

Rule 5: Friction losses change with flow

For turbulent flow, approximately:

$$h_f \propto Q^2$$

If the total flow-dependent loss is 29 m at 200 m³/h, then at 100 m³/h:

$$h_{f,100} = 29 \left(\frac{100}{200} \right)^2$$
$$h_{f,100} = 29(0.25)$$
$$h_{f,100} = 7.25 \text{ m}$$

Static elevation and vessel-pressure head usually remain approximately constant, while friction and equipment losses change with flow.

5. System Curve and Required Shutoff Rise

The approximate system curve is:

$$H_{\text{system}} = H_{\text{static}} + KQ^2$$

Classify the system as:

Static head dominated

More than 50% of the required head comes from static resistance, such as:

- Elevation difference
- Pressure difference between vessels

Friction dominated

More than 50% comes from:

- Piping friction
- Valves
- Fittings
- Heat exchangers
- Other flowing-system losses

This classification affects the required rise from rated head to shutoff.

Classify the system character when specifying the minimum rise from rated head to shutoff.

Operating condition	Minimum head rise to shutoff
Pumps operating in parallel	10%
Main firewater and jockey pumps	10%
Individual continuous pump, static-dominated system	7%
Individual continuous pump, friction-dominated system	6%
Individual intermittent pump	6%

Calculate head rise as:

$$\% \text{ head rise} = \frac{H_{\text{shutoff}} - H_{\text{rated}}}{H_{\text{rated}}} \times 100$$

Example:

Given process data

Assume a cooling-water transfer pump has:

Parameter	Value
Rated flow, Q_r	300 m ³ /h
Static head	20 m
Piping friction loss at rated flow	30 m
Heat-exchanger pressure drop at rated flow	8 m
Control-valve pressure drop at rated flow	2 m
Operating mode	Continuous
Number of operating pumps	One

The total flow-dependent loss at rated flow is:

$$H_{\text{dynamic,r}} = 30 + 8 + 2$$
$$H_{\text{dynamic,r}} = 40 \text{ m}$$

Therefore, the required rated head is:

$$H_r = H_{\text{static}} + H_{\text{dynamic,r}}$$
$$H_r = 20 + 40$$
$$H_r = 60 \text{ m}$$

Determine whether the system is static or friction dominated

Classify the system according to whether more than 50% of the required head comes from static resistance or flow-dependent resistance. For an individually operated, continuous pump, the minimum required rise to shutoff is 7% for a static-dominated system and 6% for a friction-dominated system.

Calculate the static-head percentage:

$$\text{Static percentage} = \frac{H_{\text{static}}}{H_r} \times 100$$
$$= \frac{20}{60} \times 100$$
$$\text{Static percentage} = 33.3\%$$

Calculate the dynamic-head percentage:

$$\text{Dynamic percentage} = \frac{40}{60} \times 100$$
$$\text{Dynamic percentage} = 66.7\%$$

Because the flow-dependent resistance is greater than 50%:

$$\text{The system is friction dominated}$$

Therefore, the required minimum pump head rise from rated point to shutoff is:

$$6\%$$

Develop the system-curve equation

The approximate system-curve equation is:

$$H_{\text{system}} = H_{\text{static}} + KQ^2$$

where:

- H_{system} = total system head, m
- H_{static} = static head, m
- K = system resistance coefficient
- Q = flow, m³/h

At rated flow:

$$60 = 20 + K(300)^2$$

Therefore:

$$K = \frac{60 - 20}{300^2}$$

$$K = \frac{40}{90000}$$

$$K = 0.0004444 \frac{\text{m}}{(\text{m}^3/\text{h})^2}$$

The system equation becomes:

$$H_{\text{system}} = 20 + 0.0004444Q^2$$

4. Calculate system head at several flow rates

At zero flow

$$H_{\text{system}} = 20 + 0.0004444(0)^2$$

$$H_{\text{system}} = 20 \text{ m}$$

At zero flow there are no piping-friction or equipment-flow losses. Only static head remains.

At 100 m³/h

$$H_{\text{system}} = 20 + 0.0004444(100)^2$$

$$H_{\text{system}} = 20 + 4.44$$

$$H_{\text{system}} = 24.44 \text{ m}$$

At 150 m³/h

$$H_{\text{system}} = 20 + 0.0004444(150)^2$$

$$H_{\text{system}} = 20 + 10$$

$$H_{\text{system}} = 30.0 \text{ m}$$

At 200 m³/h

$$H_{\text{system}} = 20 + 0.0004444(200)^2$$

$$H_{\text{system}} = 20 + 17.78$$

$$H_{\text{system}} = 37.78 \text{ m}$$

At rated flow, 300 m³/h

$$H_{\text{system}} = 20 + 0.0004444(300)^2$$

$$H_{\text{system}} = 20 + 40$$

$$H_{\text{system}} = 60.0 \text{ m}$$

At 120% of rated flow

$$Q = 1.20(300) = 360 \text{ m}^3/\text{h}$$

$$H_{\text{system}} = 20 + 0.0004444(360)^2$$

$$H_{\text{system}} = 20 + 57.6$$

$$H_{\text{system}} = 77.6 \text{ m}$$

The resulting system-curve points are:

Flow, m ³ /h	Static head, m	Dynamic head, m	Total system head, m
0	20.0	0.0	20.0
100	20.0	4.44	24.44
150	20.0	10.0	30.0
200	20.0	17.78	37.78
300	20.0	40.0	60.0
360	20.0	57.6	77.6

5. Calculate the required pump shutoff head

For a continuous individual pump in a friction-dominated system, the minimum rise from rated head to shutoff is 6%.

The formula is:

$$\text{Head rise} = \frac{H_{\text{shutoff}} - H_r}{H_r} \times 100$$

Rearranging:

$$H_{\text{shutoff,min}} = H_r \left(1 + \frac{\text{Required rise}}{100} \right)$$

Substitute the values:

$$H_{\text{shutoff,min}} = 60(1 + 0.06)$$

$$H_{\text{shutoff,min}} = 60(1.06)$$

$$H_{\text{shutoff,min}} = 63.6 \text{ m}$$

Therefore, the vendor's offered pump curve must have a shutoff head of at least:

$$63.6 \text{ m}$$

6. Check two vendor offers

Vendor A

Suppose Vendor A offers:

- Rated head = 60 m
- Shutoff head = 65 m

Calculate the rise:

$$\text{Rise} = \frac{65 - 60}{60} \times 100$$
$$\text{Rise} = 8.33\%$$

Comparison:

$$8.33\% > 6\%$$

Result

Vendor A satisfies the minimum rise requirement

Vendor B

Suppose Vendor B offers:

- Rated head = 60 m
- Shutoff head = 62 m

$$\text{Rise} = \frac{62 - 60}{60} \times 100$$
$$\text{Rise} = 3.33\%$$

Comparison:

$$3.33\% < 6\%$$

Result

Vendor B does not satisfy the requirement

A technical query should be issued requesting a revised hydraulic selection with a continuously rising curve and at least 6% head rise from the rated point to shutoff.

6. Select the Pump Configuration

Use the service-based selection guidance in Appendix A together with the explicit configuration restrictions in the standard. The following practical matrix supports preliminary selection, but the exact service row in the approved Appendix A remains governing.

Duty characteristic	Preliminary pump family	Engineering reason
Moderate flow and head; process service	OH2 or applicable ASME/ISO end-suction pump	Compact single-stage construction and straightforward maintenance.
High flow, moderate head	BB1	Double-suction arrangement can improve NPSH and rotor support.
One- or two-stage high-pressure process duty	BB2	Radially split between-bearings configuration.
Multistage high head	BB3 or BB5	Multiple stages; BB5 adds double-casing pressure containment.

Duty characteristic	Preliminary pump family	Engineering reason
Open wet pit or intake	VS1 or VS2	Submerged first stage and discharge through column.
Selected non-flammable oily-water sump with suction lift ≤ 6 m	Horizontal self-priming pump	Permitted where hydraulic limits and integral priming requirements are met.
Highly hazardous clean liquid	Sealless or Arrangement 3 sealed pump	Improved leakage containment.
Varying flow and head in friction-dominated system	Variable-speed pump	Reduces throttling and energy loss when economically justified.

6.1 Important overhung-pump restriction

A horizontal single-stage overhung pump is not acceptable when either condition is exceeded:

- Driver nameplate rating greater than 225 kW.
- Rated flow greater than 600 m³/h.

Selection consequence

If either limit is exceeded, evaluate a between-bearings pump rather than forcing an overhung selection.

6.2 Open-sump selection

- Stormwater open-sump installations generally use discharge-through-column vertical designs such as VS1 or VS2.
- For hydrocarbon or oily-water open sumps where the standard requires discharge through the column, separate-discharge VS4/VS5 arrangements are not acceptable.
- A direct-drive self-priming pump may be considered for applicable non-flammable oily-water service when suction lift does not exceed 6 m.
- Self-priming pumps are not used for flammable hydrocarbon service.

7. Verify BEP Location and Operating Range

7.1 API Pump

For pumps complying with API 610

- Dominant normal flow must not be less than 80% of BEP.
- Lowest normal flow must not be less than 70% of BEP.

$$Q_{\text{dominant, normal}} \geq 0.80 Q_{\text{BEP}}$$

$$Q_{\text{lowest, normal}} \geq 0.70 Q_{\text{BEP}}$$

7.2 ASME, ISO and manufacturer-standard pumps

$$0.60 Q_{\text{BEP}} \leq Q_{\text{normal/rated}} \leq 1.10 Q_{\text{BEP}}$$

Also identify MCSF, minimum continuous thermal flow, preferred operating region and allowable operating region. The minimum-flow protection setting must protect the most restrictive applicable limit.

POR shall extend from 70% to 120% of BEP flow at rated impeller trim. Rated flow shall be between 80% and 110% of BEP flow.

7.3 Example

Dominant normal = 200 m³/h, lowest normal = 170 m³/h, vendor BEP = 220 m³/h:

$$\text{Dominant normal/BEP} = 200/220 = 90.9\%$$

$$\text{Lowest normal/BEP} = 170/220 = 77.3\%$$

Both conditions satisfy the API normal-flow criteria.

8. Calculate NPSHA and Check NPSH3

For suction from an open atmospheric vessel:

$$NPSH_A = \frac{P_{\text{atm,abs}}}{\rho g} + (z_{\text{liquid}} - z_{\text{pump}}) - h_{f,s} - \frac{P_v}{\rho g}$$

For a closed vessel:

$$NPSH_A = \frac{P_{\text{vessel,abs}}}{\rho g} + (z_{\text{liquid}} - z_{\text{pump}}) - h_{f,s} - \frac{P_v}{\rho g}$$

For a closed vessel, use the vessel absolute pressure in place of atmospheric pressure.

- Use minimum vessel pressure and minimum liquid level.
- Use maximum pumping temperature and corresponding vapor pressure.
- Include maximum suction-line and permanent-strainer pressure drop.
- Evaluate rated flow and any higher mandatory operating flow.
- Use the correct NPSH datum for the pump configuration.

NPSHA to exceed NPSH3 from minimum continuous stable capacity through 120% of rated flow, with a minimum rated-point margin of:

$$NPSH_A - NPSH_3 \geq 1.0 \text{ m}$$

For main firewater pumps, the hydraulic and NPSH evaluation extends to 150% of rated capacity.

Also check suction-specific speed. Except for integrally geared pumps, the limit is:

$$N_{ss} \leq 12,000$$

The vendor curve must also show that NPSH3 remains below NPSHA over the required operating range. For main firewater pumps, the hydraulic/NPSH check extends to 150% of rated capacity.

Reliability caution

A 1 m difference is a minimum compliance check, not proof of zero cavitation damage. High-energy, hot-liquid, variable-speed and reliability-critical services may require a larger engineering margin.

9. Special Service Screening

Viscous service

Apply Hydraulic Institute HI 9.6.7 corrections to:

- Flow
- Head
- Efficiency
- Absorbed power

The method applies to conventional radial-flow pumps handling Newtonian liquids. It should not be extended indiscriminately to axial-flow, mixed-flow, slurry or non-Newtonian services.

Highly hazardous service

Select:

- Sealless pump, or
- Pump with Arrangement 3 dual pressurized seals

Sealless pumps are generally limited by

$$\text{Motor nameplate} \leq 225 \text{ kW}$$

If the required motor exceeds this value, or a sealless pump is unsuitable, use the specified pressurized dual-seal arrangement unless an approved alternative is obtained.

Solids

Unless a higher concentration is specified, pumps are required to be designed for at least:

$$300 \text{ ppm by weight}$$

of suspended solids. Actual solids service must still be evaluated for particle size, hardness, settling characteristics and erosiveness.

Variable flow and head

Variable-speed pumps should be considered when both head and flow vary, and life-cycle economics justify the selection. They are particularly effective for friction-dominated systems where required differential head varies significantly with flow.

10. Verify Pressure Casing and Temperature

Determine the maximum discharge pressure from:

$$P_{\text{discharge,max}} = P_{\text{suction,max}} + \Delta P_{\text{shutoff,max}}$$

The shutoff differential pressure must account for:

- Maximum specified liquid density
- Maximum impeller diameter for fixed-speed applications
- Maximum applicable speed for variable-speed applications
- Maximum number of installed stages
- Start-up or flushing liquid, where applicable

MAWP shall be at least maximum discharge pressure plus 10% of maximum differential pressure and shall meet the applicable minimum rating.

11. Calculate Pump Power and Select the Driver

For SI units:

$$P_{\text{shaft}} = \frac{\rho g Q H}{1000 \eta_p}$$

For Q in m^3/h :

$$P_{\text{shaft}} = \frac{\rho Q H}{367,000 \eta_p}$$

For water:

$$P_{\text{shaft}} = \frac{Q H}{367 \eta_p}$$

Check power at:

- Rated point
- Maximum specified flow
- End of curve
- Maximum impeller diameter
- Maximum variable speed
- Maximum-density liquid
- Cold viscous start-up
- Flushing-water operation

For ASME, ISO and manufacturer-standard pumps:

Driver nameplate range	Minimum percentage of pump rated power
Below 22 kW	125%
22 to 55 kW	115%
Above 55 kW	110%

For API pumps, apply the applicable API 610, 31-SAMSS-004, motor specification and datasheet requirements rather than automatically using this table.

12. Worked Selection Example

Input	Value
Service	Hydrocarbon transfer
Dominant normal flow	200 m3/h
Rated flow	220 m3/h
Rated TDH	150 m
Density	850 kg/m3
Pump efficiency	78%
NPSHA at rated flow	7.0 m
Vendor NPSH3	5.4 m
Vendor BEP	230 m3/h
Proposed motor	110 kW

Rated-flow margin

$$220/200 = 1.10 \rightarrow \text{acceptable 10\% margin}$$

BEP position

$$\text{Normal/BEP} = 200/230 = 87.0\%$$

$$\text{Rated/BEP} = 220/230 = 95.7\%$$

Both points are favorably located for an API pump.

NPSH margin

$$\text{NPSH margin} = 7.0 - 5.4 = 1.6 \text{ m}$$

The minimum 1 m rated-point criterion is satisfied, subject to full-range review.

Shaft power

$$P_{\text{shaft}} = \frac{850 \times 220 \times 150}{367000 \times 0.78} = 98.0 \text{ kW}$$

110 kW motor provides approximately 12.2% margin at the rated point. Acceptance still requires power checks at maximum flow, maximum impeller diameter and maximum liquid density.

Preliminary pump type

Flow is below 600 m3/h and proposed motor rating is below 225 kW, so a single-stage overhung API pump may be considered. Final selection still depends on required head per stage, pressure rating, temperature, NPSH, liquid hazard, materials, seal arrangement and the complete vendor curve.

Preliminary conclusion

Consider an API OH2 pump, subject to successful hydraulic, mechanical, material, seal and vendor-curve verification.

High-flow cooling-water pump

Given data

Parameter	Value
Dominant normal flow	850 m³/h
Rated flow	930 m³/h
Rated TDH	65 m
Density	997 kg/m³
Efficiency	86%
Vendor BEP	960 m³/h
NPSHA	7.0 m
Vendor NPSH3	5.6 m
Operation	Continuous

Step 1: Rated-flow check

$$Q_{r,\text{max}} = 1.10(850) = 935 \text{ m}^3/\text{h}$$

Proposed rated flow:

$$930 < 935 \text{ m}^3/\text{h}$$

The rated-flow margin is acceptable.

Step 2: Rule out an overhung pump

The rated flow is:

$$930 \text{ m}^3/\text{h} > 600 \text{ m}^3/\text{h}$$

It does not permit a horizontal single-stage overhung pump when rated flow exceeds 600 m³/h.

Therefore:

OH2 is not acceptable

Step 3: BEP check

$$\frac{930}{960} \times 100 = 96.9\% \text{ BEP}$$

This is an excellent rated-point location.

Step 4: Power

$$P_{\text{shaft}} = \frac{997(930)(65)}{367000(0.86)}$$

$P_{\text{shaft}} = 190.8 \text{ kW}$

A motor near 220 kW or 225 kW may be considered, provided it covers maximum curve power and other applicable conditions.

Step 5: NPSH

$$NPSH_{\text{margin}} = 7.0 - 5.6 = 1.4 \text{ m}$$

This passes the minimum numerical criterion but is not generous. Check:

- Lowest source level
- Maximum cooling-water temperature
- Fouled strainer
- Suction-line pressure drop
- Maximum operating flow
- Air entrainment

Preliminary selection

API BB1, single-stage double-suction pump

Why BB1

- High flow
- Moderate head
- Better rotor support than an overhung configuration
- Double-suction impeller can reduce inlet velocity and NPSH3
- Axially split construction provides good maintenance access

High-temperature, high-pressure process pump

Given data

Parameter	Value
Service	Hot hydrocarbon charge
Dominant normal flow	300 m ³ /h
Rated flow	330 m ³ /h
Rated TDH	280 m
Density	690 kg/m ³
Temperature	285°C
Efficiency	79%
Maximum suction pressure	18 bar(g)
Shutoff head at maximum impeller	350 m
Operation	Continuous

Step 1: Rated-flow check

$$\frac{330}{300} = 1.10$$

The rated-flow margin is acceptable.

Step 2: Calculate rated shaft power

$$P_{\text{shaft}} = \frac{690(330)(280)}{367000(0.79)}$$
$$P_{\text{shaft}} = 220 \text{ kW}$$

The final motor is likely greater than 225 kW after considering maximum curve power and margin.
A horizontal single-stage overhung pump is not acceptable if the driver nameplate rating exceeds 225 kW.

Step 3: Calculate maximum discharge pressure

The pressure corresponding to maximum shutoff head is:

$$\begin{aligned}\Delta P_{\text{shutoff}} &= \rho g H \\ &= 690(9.81)(350) \\ &= 2,369,115 \text{ Pa} \\ \Delta P_{\text{shutoff}} &= 23.69 \text{ bar}\end{aligned}$$

Maximum discharge pressure :

$$\begin{aligned}P_{\text{discharge,max}} &= P_{\text{suction,max}} + \Delta P_{\text{shutoff}} \\ &= 18 + 23.69 \\ P_{\text{discharge,max}} &= 41.69 \text{ bar(g)}\end{aligned}$$

The pressure casing, flanges, seals and auxiliaries must be selected for this maximum condition, not merely the rated discharge pressure.

Preliminary selection

API BB2, one- or two-stage radially split pump

Why BB2 rather than OH2

- Higher pressure and temperature capability
- Between-bearings rotor support
- Radially split casings suitable for elevated-pressure service (Temp. more than 200 degree c which triggers radially split casing (From BB series, only BB2 and BB5 are radially split casing)
- Centerline mounting accommodates thermal expansion
- Power likely exceeds the overhung limit

When to consider BB5 instead

Use BB5 if the required head cannot be achieved efficiently in one or two stages, or if the final pressure containment and service severity justify a multistage double-casing barrel pump.

Condensate pump with low NPSHA

Given data

Parameter	Value
Service	Hot condensate
Dominant normal flow	220 m ³ /h
Rated flow	240 m ³ /h
Rated head	160 m
Density	950 kg/m ³
Temperature	95°C
NPSHA	2.8 m
OH2 vendor NPSH3	4.2 m
VS6 vendor NPSH3	1.6 m
Efficiency of proposed VS6	78%

Step 1: Check OH2 NPSH

$$NPSH_{\text{margin}} = 2.8 - 4.2 = -1.4 \text{ m}$$

The available NPSH is lower than NPSH3.

OH2 selection fails

Increasing the motor size or casing pressure rating will not solve this problem because the failure is on the suction side.

Step 2: Possible solutions

The engineer could:

1. Lower the horizontal pump elevation.
2. Increase condensate-vessel liquid level.
3. Increase vessel pressure.
4. Reduce suction losses.
5. Reduce pump speed.
6. Use a lower-NPSH first-stage design.
7. Select a vertical double-casing pump with the first-stage impeller below the mounting level.

Step 3: Check VS6 proposal

$$NPSH_{\text{margin}} = 2.8 - 1.6 = 1.2 \text{ m}$$

The minimum 1 m rated-point criterion is satisfied, subject to checking the complete operating range.

Step 4: Power

$$P_{\text{shaft}} = \frac{950(240)(160)}{367000(0.78)}$$

$P_{\text{shaft}} = 127.5 \text{ kW}$

Preliminary selection

API VS6 vertical double-casing pump

Why

- Low NPSHA is the governing constraint.
- The first-stage impeller can be positioned lower.
- Additional static submergence increases pressure at the impeller eye.
- The multistage vertical configuration can develop the required head.

Why not the other VS series?

The correct choice depends primarily on the **source arrangement, hydraulic duty, pressure containment and pump construction**, not only NPSHA.

VS1

A VS1 is a wet-pit, vertically suspended, single-casing diffuser pump with discharge through the column.

VS1 may be appropriate when:

- Condensate is collected in an open hotwell or pit.
- The bowl assembly is submerged directly in the liquid.
- The liquid surrounding the bowl provides suction submergence.
- The required head can be generated by the selected number of diffuser stages.
- A pressure-containing outer can is unnecessary.
- Discharge-through-column construction is acceptable.

Why VS1 could be a better candidate

If this is a conventional open condensate hotwell, VS1 may solve the low-NPSH problem simply by locating the first-stage bowl below the minimum liquid level.

In that case, an outer VS6 pressure can may add unnecessary:

- Cost
- Weight
- Fabrication complexity
- Foundation depth
- Pressure-boundary inspection
- Maintenance complexity

Therefore:

Open hotwell or wet pit: VS1 should be evaluated before assuming VS6

VS2

A VS2 is a wet-pit, vertically suspended, single-casing, double-volute pump with discharge through the column.

VS2 is generally considered when:

- The pump is installed in a wet pit.

- Relatively high flow is required.
- Volute hydraulics are more appropriate than diffuser hydraulics.
- The first stage remains submerged.

For a 240 m³/h condensate duty, VS2 may be technically possible, but the required head of 160 m typically points toward multistage diffuser hydraulics rather than a high-flow double-volute arrangement.

Therefore, VS2 is less likely than VS1 or VS6 for the assumed example, but it cannot be rejected solely from the limited information provided.

VS3

A VS3 is a wet-pit, vertically suspended axial-flow pump with discharge through the column.

Axial-flow pumps are used for:

- Very high flow
- Very low head
- Circulation
- Flood control
- Cooling-water intake
- Drainage

The example requires:

$$H = 160 \text{ m}$$

That is completely inconsistent with the normal hydraulic application of an axial-flow VS3 pump.

Therefore:

VS3 is unsuitable because the required head is too high

VS4

A VS4 is a vertically suspended, volute, line-shaft-driven sump pump with a separate discharge pipe.

VS4 is generally used for:

- Sump drainage
- Oily water
- Process sumps
- Moderate flow and head
- Applications where separate discharge piping is acceptable

Why VS4 is less attractive for example

The example requires:

$$H = 160 \text{ m}$$

A conventional VS4 sump pump is not normally the first choice for this relatively high head. Also, a VS4 is a single-casing sump arrangement rather than a pressure-can multistage condensate service pump.

It may be considered only if a vendor has a technically proven multistage or special design appropriate for the actual duty and project specification.

VS5

A VS5 is a vertically suspended cantilever sump pump.

It normally has:

- No submerged line-shaft bearings
- Limited shaft length
- A cantilever rotor
- Separate discharge piping
- Moderate or low head capability

For:

$$Q = 240 \text{ m}^3/\text{h}$$

and:

$$H = 160 \text{ m}$$

a VS5 would generally be unsuitable because:

- The head is too high for a typical cantilever sump-pump application.
- Shaft stiffness and overhang become limiting.
- It is not normally selected as a high-head multistage condensate pump.

Therefore:

VS5 is not a reasonable preliminary choice
--

VS6

VS6 is a double-casing, diffuser, vertically suspended pump.

VS6 becomes attractive when:

- Low NPSHA requires lowering the first-stage impeller.
- The pump is not simply suspended in an open wet pit.
- A suction can or barrel is needed.
- The service requires several diffuser stages.
- The discharge pressure is relatively high.
- The pump is fed from a closed vessel or suction piping.
- Pressure-containing construction around the bowl and column is required.
- The pump must develop substantial head.

For the assumed 160 m head, the diffuser-type multistage arrangement makes VS6 a credible candidate.

Oily-water sump with 8 m suction lift

Given data

Parameter	Value
Service	Non-flammable oily water
Sump	Open
Suction lift	8 m
Rated flow	60 m ³ /h
Rated head	32 m
Solids	700 ppm
Operation	Intermittent

Step 1: Check self-priming limit

A horizontal direct-drive self-priming pump for applicable oily-water open-sump service when suction lift does not exceed 6 m.

Here:

8 m > 6 m

Therefore:

Horizontal self-priming pump is not acceptable

Step 2: Select a submerged arrangement

A vertically suspended pump eliminates suction lift if the first-stage impeller remains submerged below the minimum liquid level.

Preliminary selection

VS1 or VS2, discharge through column

Why not VS4 or VS5 automatically?

For oily-water and hydrocarbon open-sump applications covered by the relevant requirement, discharge-through-column constructions are preferred or required, while separate-discharge configurations may be prohibited.

Additional design checks

- Minimum submergence
- Vortex prevention
- Sump geometry
- Solids passage
- Nonmetallic bushing compatibility
- Minimum liquid level
- Shaft length and critical speed
- Starting frequency
- Mechanical-seal or discharge-head sealing arrangement
- Strainer clearance from sump bottom

Highly hazardous chemical service

Given data

Parameter	Value
Dominant normal flow	45 m³/h
Rated flow	49 m³/h
Rated head	85 m
Density	1,150 kg/m³
Viscosity	4 cP
Pump efficiency	60%
Solids	Negligible
Motor estimate	Below 30 kW
Hazard classification	Highly hazardous

Step 1: Calculate shaft power

$$P_{\text{shaft}} = \frac{1150(49)(85)}{367000(0.60)}$$

$P_{\text{shaft}} = 21.8 \text{ kW}$

Step 2: Determine leakage-control philosophy

A sealless pump or a pump with Arrangement 3 dual pressurized seals for highly hazardous services, subject to the stated applicability requirements. Sealless pumps are limited to a motor nameplate rating of 225 kW.

The estimated motor is well below 225 kW, so a sealless pump is feasible from a power-limit perspective.

Option A: Canned-motor pump

Suitable when:

- The liquid provides acceptable bearing lubrication.
- Motor heat can be removed.
- Solids are negligible.
- Internal circulation paths will remain clean.
- Leakage containment is critical.

Option B: Magnetic-drive pump

Suitable when:

- Process isolation is required.
- Containment-shell losses and temperature are acceptable.
- Magnetic coupling has adequate torque margin.
- The liquid does not crystallize or polymerize in internal passages.

Option C: API pump with Arrangement 3 seals

Prefer when:

- Sealless-pump internal lubrication is unsuitable.
- Solids or viscosity create internal circulation concerns.
- Higher temperature or pressure makes the sealless option impractical.
- A reliable barrier-fluid system is available.

Preliminary selection

For clean and lubricating liquid:

Canned-motor or magnetic-drive sealless pump

For poor lubricity, crystallization or difficult internal circulation:

API OH2 with Arrangement 3 dual pressurized seals

Wide-flow-range circulation service

Given operating cases

Case	Flow	Required head
Minimum	180 m ³ /h	25 m
Normal	360 m ³ /h	42 m
Maximum	520 m ³ /h	68 m

The system is predominantly friction controlled.

A fixed-speed pump is offered at:

- Rated flow: 520 m³/h
- Rated head: 68 m
- BEP: 540 m³/h
- Rated speed: 1,480 r/min

Problem with fixed-speed selection

At minimum and normal flow, a control valve would have to dissipate substantial excess head. The pump could also spend considerable time well to the left of BEP.

At minimum flow:

$$\frac{180}{540} \times 100 = 33.3\% \text{ BEP}$$

This is far below the normal operating range expected for an API/SAMSS pump.
Therefore, selecting one fixed-speed pump solely for the maximum case is unsuitable.

Alternative 1: Variable-speed pump

Using the approximate head affinity law for the normal point:

$$\frac{H_2}{H_1} = \left(\frac{N_2}{N_1}\right)^2$$
$$N_2 = 1480 \sqrt{\frac{42}{68}}$$

$N_2 \approx 1163 \text{ r/min}$

This is preliminary only. The operating point must be confirmed from the full reduced-speed curves and system curve.
Aramco standard recommends considering variable-speed pumps for varying head and flow when economically justified, particularly in friction-dominated systems.

Alternative 2: Multiple smaller pumps

Use three 50% pumps:

- Two operating at maximum demand
- One operating near lower demand
- One standby, depending on availability philosophy

This may keep individual pumps closer to BEP across the operating range.

Preliminary selection

Variable-speed pump or multiple smaller parallel pumps

The final choice requires:

- Life-cycle energy comparison
- Minimum continuous speed
- Motor cooling at low speed
- Critical-speed separation
- NPSH3 at maximum speed
- Availability philosophy
- Parallel-curve compatibility
- Minimum-flow protection

Multistage high-head water-injection pump

Given data

Parameter	Value
Rated flow	500 m³/h
Rated head	1,800 m
Density	1,020 kg/m³

Parameter	Value
Efficiency	82%
Proposed head per stage	225 m
Maximum suction pressure	12 bar(g)
Operation	Continuous

Step 1: Preliminary number of stages

$$n = \frac{H_{\text{total}}}{H_{\text{stage}}}$$

$$n = \frac{1800}{225}$$

$n = 8 \text{ stages}$

The final number of stages must come from actual vendor hydraulics, not only arithmetic division.

Step 2: Shaft power

$$P_{\text{shaft}} = \frac{1020(500)(1800)}{367000(0.82)}$$

$P_{\text{shaft}} \approx 3050 \text{ kW}$

Step 3: Configuration screening

An overhung pump is impossible because:

- Power greatly exceeds 225 kW.
- The duty requires multiple stages.
- Discharge pressure is very high.

BB1 and BB2 are normally unsuitable because the required head demands several stages.

Step 4: Preliminary type

API BB5 multistage barrel pump

Why BB5

- High total head
- High discharge pressure
- Multistage construction
- Double pressure casing
- Cartridge-type inner assembly can support maintainability
- Suitable for severe continuous high-energy duty